

Manuel Magana

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EDUCATION

Northeastern University

M.S. Computer Science

San Jose, CA

May 2026

Relevant Courses: Algorithms, Distributed Systems, Computer Networking, Software Engineering, Machine Learning, Deep Learning

University of California, Berkeley

B.A. Molecular and Cell Biology: Neurobiology

Berkeley, CA

May 2021

Relevant Courses: Data Structures, Machine Structure, Principles & Techniques of Data Science

PROJECTS

WhatToDo | *Python, FastAPI, Docker, AWS, PostgreSQL* | [GitHub](#)

Jan 2026 – Present

- Drafted and owned backend design for a 4-person team: a 3-model SQLAlchemy schema on AWS RDS with JWT auth and bcrypt password hashing.
- Built a dual-provider AI engine using an abstract base class for provider-agnostic discovery with automatic failover between LLM providers.
- Established CI/CD pipeline with GitHub Actions and deployed containerized API to AWS (App Runner, ECR).

Rate Limiter Service | *Go, Redis, Docker* | [GitHub](#)

Jan 2026

- Built a Go reverse proxy implementing fixed window and token bucket rate limiting with configurable per-user quotas.
- Used Redis transactions for atomic token consumption to prevent race conditions; validated under 1,000+ concurrent requests.

Paxos Key-Value Store | *Java, RPC* | [GitHub](#)

Apr 2025

- Implemented a fault-tolerant distributed KV store using Paxos consensus with Proposer, Acceptor, and Learner roles over RPC.
- Validated under chaos testing: 10% acceptor failure rate, leader crashes, and network partitions while maintaining consensus.

CPU Scheduler Simulator | *C* | [GitHub](#)

Nov 2025

- Built a process scheduler implementing MLFQ, Round Robin, and SJF with preemption, priority boosting, and I/O wait handling via custom min-heap queues.

Signal Analytics API | *Python, FastAPI, DuckDB* | [GitHub](#)

Sept 2025 – Jan 2026

- Architected a two-layer system: a Python ETL library and a FastAPI REST API with DuckDB, offset pagination, composable filters, and Pydantic validation.

EXPERIENCE

Genentech

Bioinformatics Scientist

San Francisco, CA

Nov 2022 – Jul 2025

- Co-designed a Python validation library with 2 engineers, writing unit tests and conducting PR reviews; wrote scripts consuming input files via the library and containerized them with Docker for an internal workflow system used by 20+ users.
- Architected a coordinator-worker batch system with task queuing and concurrent execution, enabling parallel processing of large-scale analysis jobs.
- Built a PySpark data pipeline to process a 200GB+ dataset that exceeded existing tooling, using partition tuning and Parquet for efficient filtering and analysis.

Endless West

Junior Data Scientist

San Francisco, CA

Mar 2021 – Nov 2021

- Developed Python web scrapers to extract and structure data from public databases, replacing hours of manual lookup with a searchable local dataset.

TECHNICAL SKILLS

Languages: Python, Go, Java, C, SQL, R, Bash

Frameworks & Tools: FastAPI, pytest, JUnit, Docker, AWS, Git, Redis, PostgreSQL, DuckDB, GitHub Actions, CI/CD

Libraries: pandas, PySpark, scikit-learn, PyTorch, TensorFlow/Keras